1 Introduction

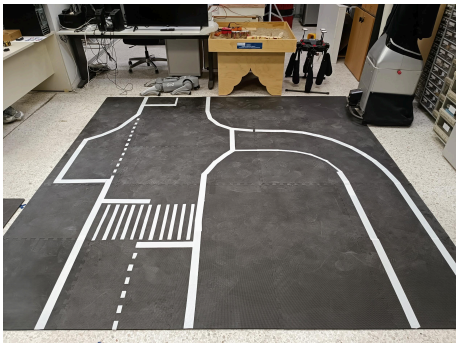
In this phase, our team focused on refining perception and control, integrating sensor fusion for precise odometry, improving traffic sign and lane detection, and expanding the test track.

2 Planned activities

- Expand the test track.
- 3D Print final form of the traffic signs.
- Integration of Magnetic Wheel Encoders and Sensor Fusion for Precise Odometry
- Refine the sign and lane detection models and merge them on the car's pipeline.
- Extensively tested the autonomy level of the vehicle.

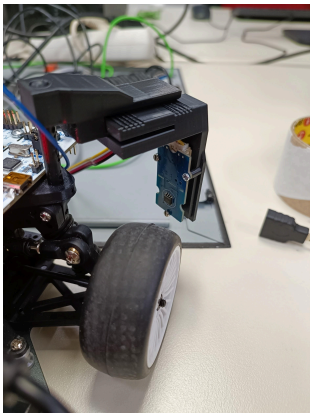
3 Status of planned activities

➔ **Expand the test track.**
Status: **Completed**
➔ **Implementation:** Added a roundabout section and refined the already existing one.



➔ **Creating the traffic signs.**
Status: **Completed**
Implementation: Using the provided models for the traffic signs, we 3D printed them, according to the exact measures specified.

➔ **Sensor Fusion for Precise Odometry.**
Status: **Completed**
Implementation: Our autonomous vehicle now features precision wheel encoders using AS5600 magnetic sensors integrated with the STM32 Nucleo microcontroller. We developed dedicated software components for reliable, real-time sensor reading, data streaming, and accurate odometry calculations. Combined with the IMU via an Extended Kalman Filter, this system provides drift-free, millimeter-level localization. Custom-designed, 3D-printed



mounts securely position the sensors for optimal performance and stability.

➔ **Traffic sign detection.**
Status: **Completed**
Implementation: Re-trained current model on a bigger dataset that included the 3D printed signs acquired from the car's own camera. Developed state system, with different movement behaviours assigned to each detected sign.

➔ **Lane detection algorithm improvement.**
Status: **On Progress**
Implementation: Sliding window algorithm was discarded and replaced with a lightweight opencv approach in which the image becomes binary form and using Hough transforms detects straight lines. The turning offset is calculated based on the middle of the detected lines and the center of the camera.

Complications:

- Issues with merging the sign detection algorithm with the lane detection algorithm due to connectivity. (two processes requiring camera feedback simultaneously)
- Detecting curved road lanes was challenging because of limited camera fov.
- Battery degradation, not enough for the 5 minute video. We aim to acquire a new one for future experimental endeavors.

4 General status of the project

The car detects all traffic signs and calculates the distance between the car and each sign. It successfully reacts to traffic signs detected, like speeding up on highway entry, slowing down on highway exits and crosswalks and stopping completely when it detects a stop sign. The lane detection algorithm is giving a control command to the vehicle, based on the detected sign or lane. Finally, due to battery problems the car cannot be tested on our track.

5 Upcoming activities

- Reduce computational cost of each algorithm.
- Enhance robustness of lane detection algorithm.
- Give control commands to the vehicle based on detected sign and lanes.